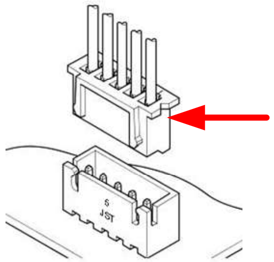
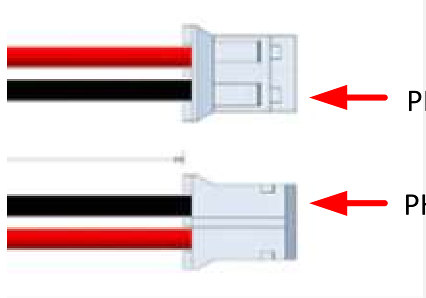
Buffer Connections

Items required for the buffer
1 x Mellow LLL Buffer
2 x JST PHR-2
1 x JST XH 3-Way
1 x JST XH 4-Way

Control Board connections required
3 inputs,, Filament Detect , Button control Load/unload
2 outputs, control buffer load/unload



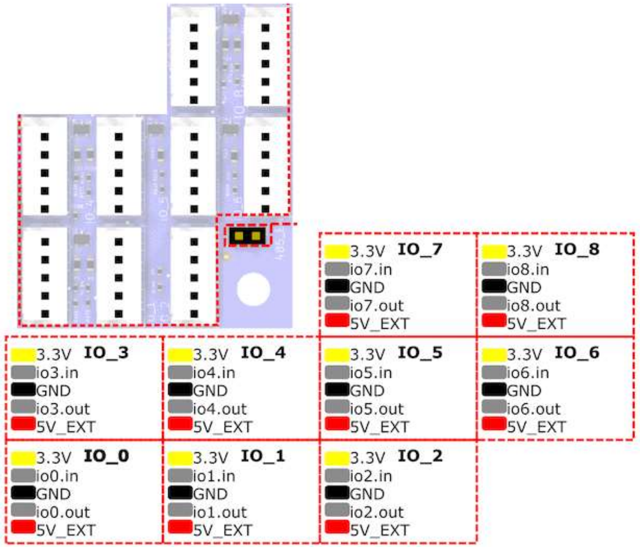
XH Notch Pin 1



PH Pin 1

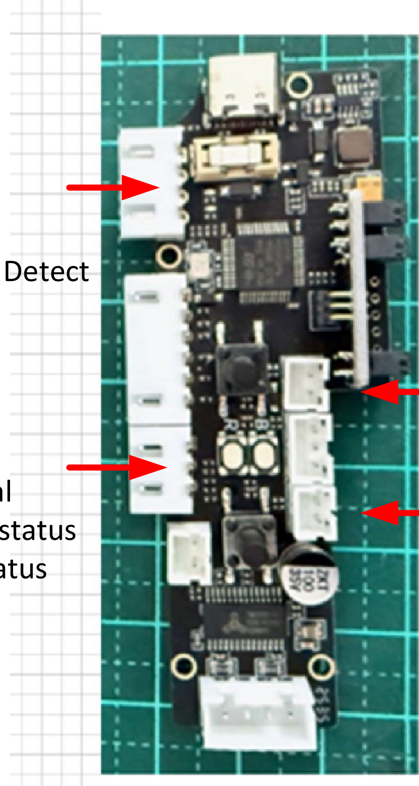
PH Pin 1

Note:- Twist the supply wires together



Example of connections to a Duet 6HC

A1)12/24
A2)0V
A3)Filament Detect
A4)N/C



C1) 0V
C2) Filament Feed

B1)0V signal
B2)Retract status
B3)Feed Status

D1) 0V
D2) Filament Retract

Con/Pin	From Buffer	Operation	wire Col/Size	TO
A1	12/24V	Positive supply	Red/22	Positive 12/24V Supply
A2	0V	0V supply	Black/22	Negative DC supply
A3	Filament Detect	Filament Detect Status	White/22	IO_8_in
B1	0V signal	Gnd Signal	Black/22	IO_7_GND
B2	Retract Status	Filament Retract Signal	Yellow/22	IO_6_in
B3	Feed Status	Filament Feed Signal	Blue/22	IO_7_in
C2	Filament Feed	Filament Buffer Feed	Green/24	IO_8_out
D2	Filament Retract	Filament Buffer Retract	White /24	IO_7_out

JST XH crimp wire size 22 > 26AWG
JST PH crimp wire size 24 > 30AWG
Stripping Length: 1.9 – 2.5 mm.

Working Principle

When the mainboard's GPIO pin outputs a low-level signal, the buffer detects this signal and performs the corresponding action:

Buffer Pin	Trigger Signal	Action Performed	Duration
Feed Control	Low Level	Buffer continuously feeds filament	Executes while the signal is held low
Retract Control	Low Level	Buffer continuously retracts filament	Executes while the signal is held low

Button	Operation	Signal Output (Buffer Pin)	Signal Type	Duration
Feed Button	Click	FEED pin outputs a high-level pulse	High Level	Automatically returns to low level after 3 seconds
Feed Button	Press & Hold	Continuous feeding	High Level	Until the button is released
Retract Button	Click	RETRACT pin outputs a low-level pulse	Low Level	Automatically returns to high level after 3 seconds
Retract Button	Press & Hold	Continuous retraction	Low Level	Until the button is released

Note: The buffer stops the action when the signal returns to a high level.

Working Operation Buffer Buttons

Click retract button , will read the current filament loaded into DWC, heat the Hotend to the correct temperature and retract the filament .

ct

Click the Load button , This will just load the filament until the filament stops automatically,

To Cancel filament feeding press retract button for >1 second

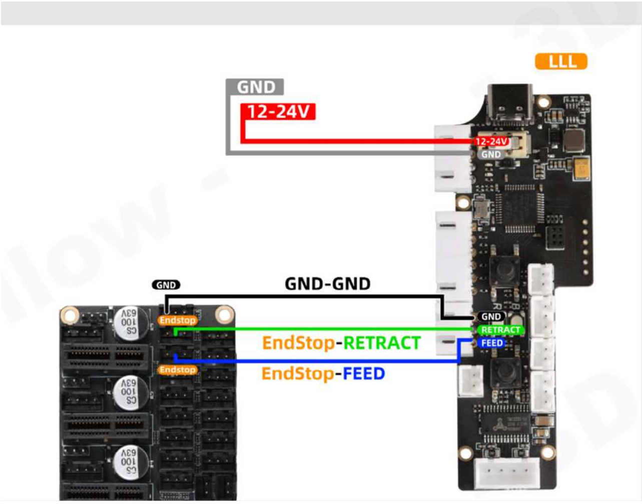
Press and hold the Retract button , **Will just force the buffer to retract the filament , make sure the filament is not still in the extruder!**

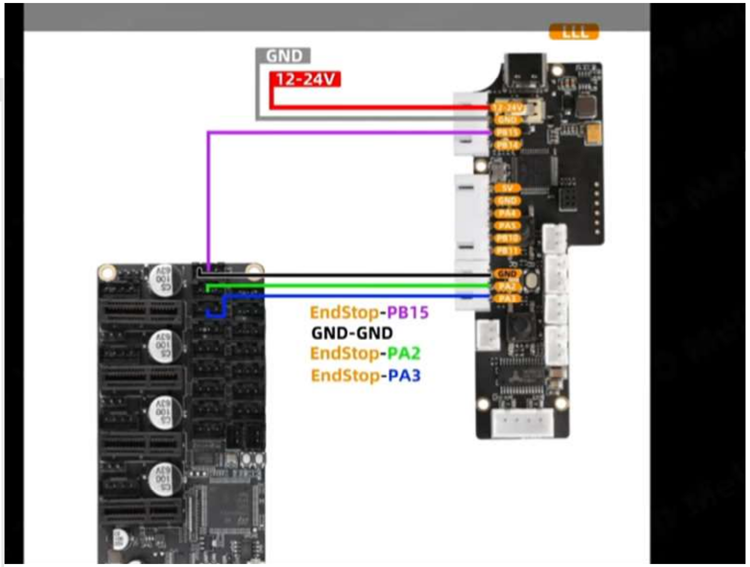
Press and hold the Load button, **will force the filament to feed until the filament starts skipping in the buffer, not recommended!**

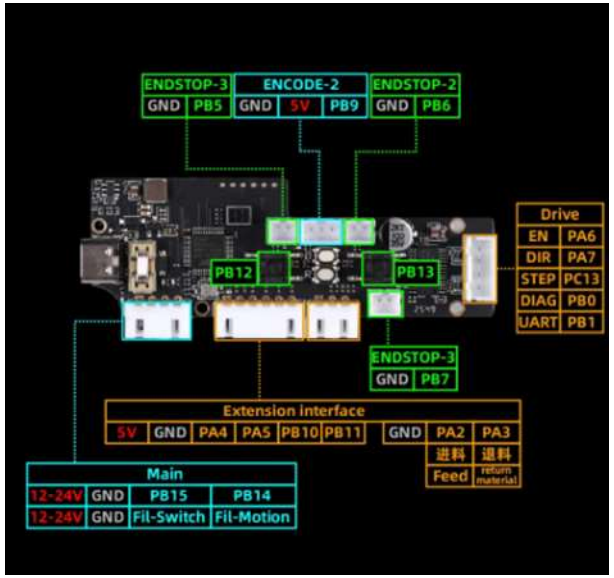
How to up date Firmware

https://mellow.klipper.cn/en/docs/ProductDoc/ExtensionBoard/fly-buffer-plus/update

https://drive.google.com/drive/folders/1wjAlmvND2FCBo72GI6J4LZEO3XSbTHVE







Signal Output Description

Button	Operation	Signal Output (Buffer Pin)	Signal Type	Duration
Feed Button (FEED)	Click	FEED pin outputs a high-level pulse	High Level	Automatically returns to low level after 3 seconds
Feed Button (FEED)	Long Press	Continuous feeding	High Level	Until the button is released
Retract Button (RETRACT)	Click	RETRACT pin outputs a low-level pulse	Low Level	Automatically returns to high level after 3 seconds
Retract Button (RETRACT)	Long Press	Continuous retraction	Low Level	Until the button is released

Working Principle

When the mainboard's GPIO pin outputs a low-level signal, the buffer detects this signal and performs the corresponding action:

Buffer Pin	Trigger Signal	Action Performed	Duration
PB5	Low Level	Buffer continuously feeds filament	Executes while the signal is held low
PB6	Low Level	Buffer continuously retracts filament	Executes while the signal is held low

Note: The buffer stops the action when the signal returns to a high level.